GEMSTAR | TASK 24

Three-Sensor Association and Exploratory EKF Tracking

Research findings, methods, evidence audit, limitations and contributions to an energy-aware ARM-FPGA perception thesis

Report type: Task-end technical research paper / evidence synthesis
Evidence scope: one clean V2 moving-target development recording, 30 original MATLAB figures, source-level exporter inspections and read-only OSI audits
Prepared: 9 November 2026

Abstract

This paper documents the experimental development and validation of a Camera-LiDAR-Radar fusion pipeline in GEMSTAR, centered on geometric association and extended Kalman filter (EKF) tracking of a moving target. The experiment sampled 53 frames from a synchronized 2,122-frame SensorView/GroundTruth recording, established a provisional LiDAR Track 2, admitted four exploratory Radar velocity updates, and evaluated one low-confidence Camera bearing hypothesis. An independent 23-frame target correspondence audit identified GT object 76897 as the nearest object to Track 2 in all evaluated frames. Mean Euclidean position error relative to the exported GT center changed from 1.650417 m (LiDAR-only) to 1.628205 m (LiDAR-Radar) and 1.628213 m (three-sensor exploratory variant). This small apparent benefit is strongly conditioned by a LiDAR-surface/GT-center reference mismatch. Source and full-rate data audits also exposed world-axis-aligned bounding-box half-extents in the GT dimension fields and severe GT velocity-reference disagreement. The experiment establishes an auditable fusion and evaluation workflow, not a demonstrated energy-efficient or statistically validated three-sensor production system.

Key conclusions

Finding	Evidence	Interpretation
Moving target correspondence	23/23 nearest-GT matches	Strong evaluation correspondence; no GT-assisted EKF updates
Position comparison	1.650417 / 1.628205 / 1.628213 m	Radar improvement about 2.22 cm; Camera benefit not demonstrated
Velocity GT integrity	1.990 median active-speed ratio; 883 zero flags	Exported actor velocity is not a validated physical reference
Thesis readiness	Reproducible offline association and tracking	Foundation for Task 25 reliability and later CPU/FPGA measurements

Scientific status: exploratory single-sequence development study. Accuracy, safety and energy guarantees have not been demonstrated.

1. Research question and thesis context

The thesis aims to minimize measured total energy per valid perception result while preserving predefined perception quality and tail-latency constraints. Task 24 provides a perception-function baseline: can asynchronous and imperfect Camera, LiDAR and Radar observations be associated without using Ground Truth inside the estimator, and can the effects of each modality be distinguished experimentally? This baseline is required before comparing ARM CPU and FPGA execution fairly.

Task 24 objectives

Build interpretable target hypotheses, associate complementary sensor observations using explicit geometric assumptions, track an estimated target state using an EKF, and assess output quality using Ground Truth only in post-processing. Keep measurements immutable and preserve the provenance of acceptance/rejection decisions.

Relation to the task roadmap

Roadmap stage	Purpose	How Task 24 contributes
Task 23	Camera/LiDAR/Radar preprocessing readiness	Provides measurement extraction, nominal coordinates and data checks
Task 24	Association and EKF tracking	Creates provisional tracks, association evidence and fusion output
Task 25	Reliability and fusion continuation	Needs observability, confidence, validity and uncertainty features
Task 26-29	CPU profiling/design-space selection	Requires frozen measurable fusion kernels
Task 30-41	ARM-FPGA acceleration, energy-aware optimization	Must preserve matched perception quality and measure system energy
Task 42-50	Live integration and validation	Must reproduce decision traces and benchmark against fair baselines

2. Dataset and experimental controls

Property	Value / method
Recording	Clean C02 V2 SensorView and GroundTruth, 2,122 matched message frames
Sampled analysis	53 frames; 172,839 LiDAR XYZ points; 3,348 Radar returns
Provisional object tracks	7 track hypotheses; Track 2 analyzed as moving target; Track 7 approximately stationary
Moving Track 2	24 sampled positions, source indices 0-938; 23 valid GT comparisons
Radar	120 provisional Track 2 associated returns in 12/24 track frames; 22 nonzero-Doppler returns across five frames
Camera	One low-confidence box at frame 530 (truck|boat, confidence 0.2105), used for exploratory bearing only
GT separation	Ground Truth used for evaluation and source audits; never as EKF input

Because the dataset is one development sequence, all comparisons are within-sequence and temporally correlated. The available results do not estimate generalization to rain, fog, traffic variations or other target classes.

3. Experimental pipeline: what was implemented and how

3.1 Data extraction and synchronization

OSI SensorView and GroundTruth streams were parsed and cross-checked using original frame indices and timestamps. Sampled LiDAR points and Radar returns were converted into vehicle coordinates. Existing x_vehicle_m/y_vehicle_m/z_vehicle_m columns already include the nominal sensor transformations; mounting offsets were not reapplied during fusion. The GT target was identified independently during evaluation.

3.2 LiDAR preprocessing and provisional tracks

LiDAR returns were filtered and grouped into local geometric clusters. Candidate positions and temporal continuity produced provisional tracks. Track 2 is the moving-target case study; the estimator is anchored to a LiDAR-observed surface/cluster reference, not an independently calibrated vehicle centroid. The LiDAR-only EKF acts as the matched baseline.

3.3 Radar association and Doppler measurement

Radar returns were associated provisionally using spatial compatibility with the LiDAR target. Only five sampled frames yielded nonzero-Doppler information for Track 2: 245, 286, 326, 775 and 816. The Radar-aided EKF accepted four exploratory velocity updates (245, 286, 326 and 816); gating was based on an innovation criterion with threshold 9. Radar measurement noise was conservatively assumed rather than calibrated. Frame 530 had no Track 2 Radar association, despite other returns in the scene.

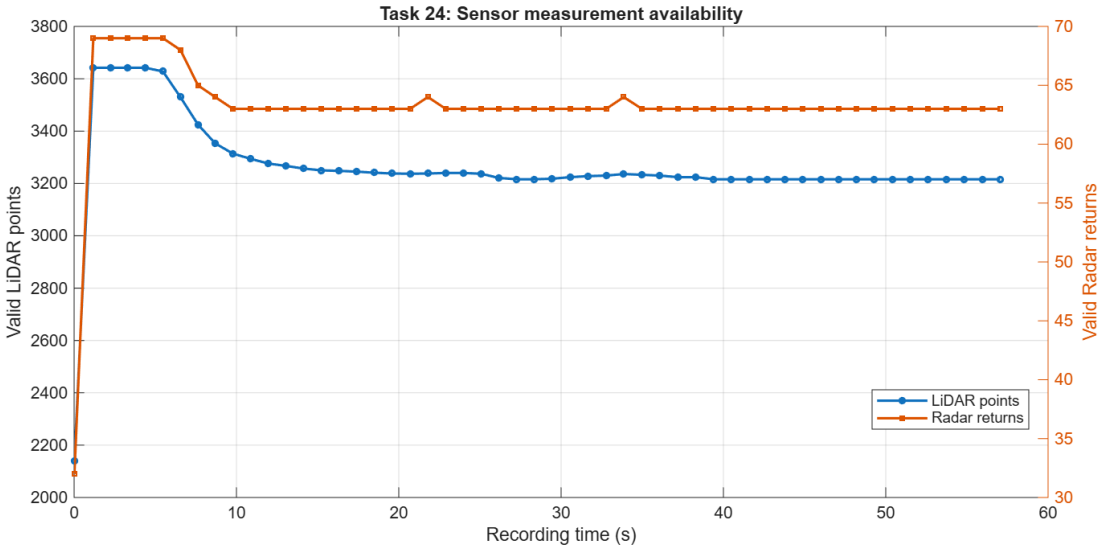
3.4 Camera hypothesis and nominal projection

A 640 x 480 image and nominal 90-degree horizontal field of view were used with fx=fy=320 px, cx=320, cy=240. Under nominal camera pose, 56 projected LiDAR points overlapped the frame-530 detection box. Approximately 51 were concentrated near a selected 20-22 m LiDAR-forward-depth interval, but points from the scene/background were also present. One tentative Camera bearing update was admitted experimentally. This does not establish calibrated Camera target identity or successful simultaneous three-way association.

3.5 EKF estimates and evaluation

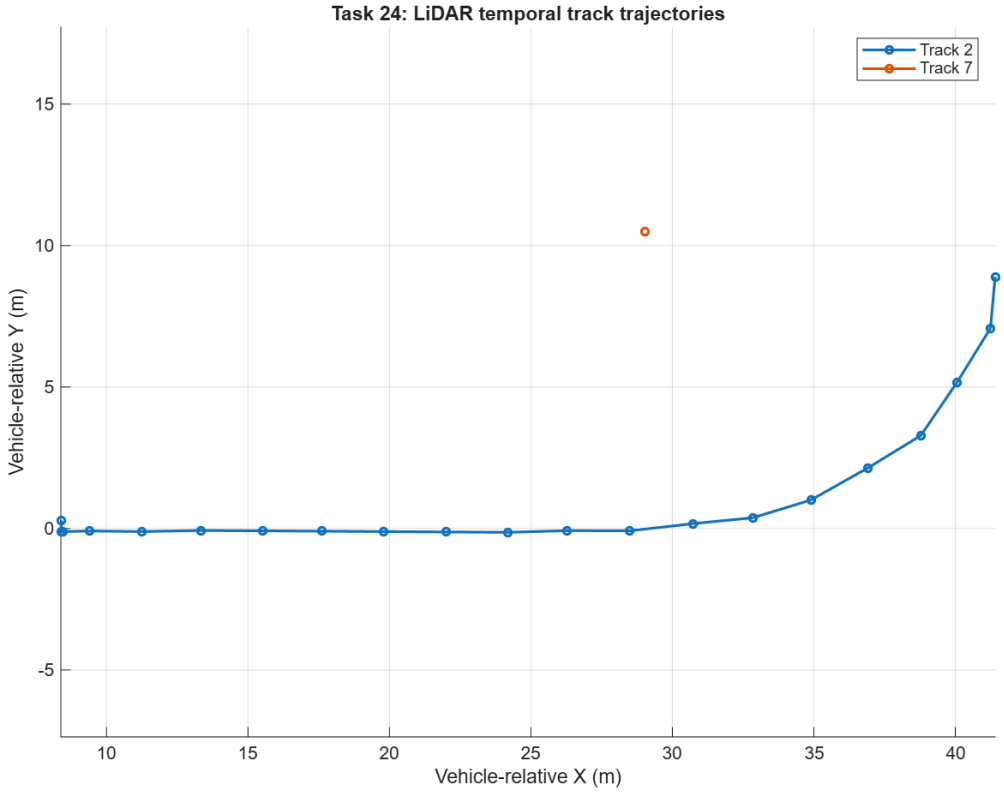
The LiDAR-only, LiDAR-Radar and exploratory three-sensor variants were compared on the same sampled recording. GT actor/world position was transformed into the ego frame for post hoc evaluation. For 23 evaluated frames, independent nearest-center matching favored GT object 76897, model BP_Small_Car_C; no GT-derived positions or identities were injected into original tracking updates.

4. Sensor availability and track-building evidence



Original MATLAB Figure 01. LiDAR and Radar measurement availability across sampled recording.

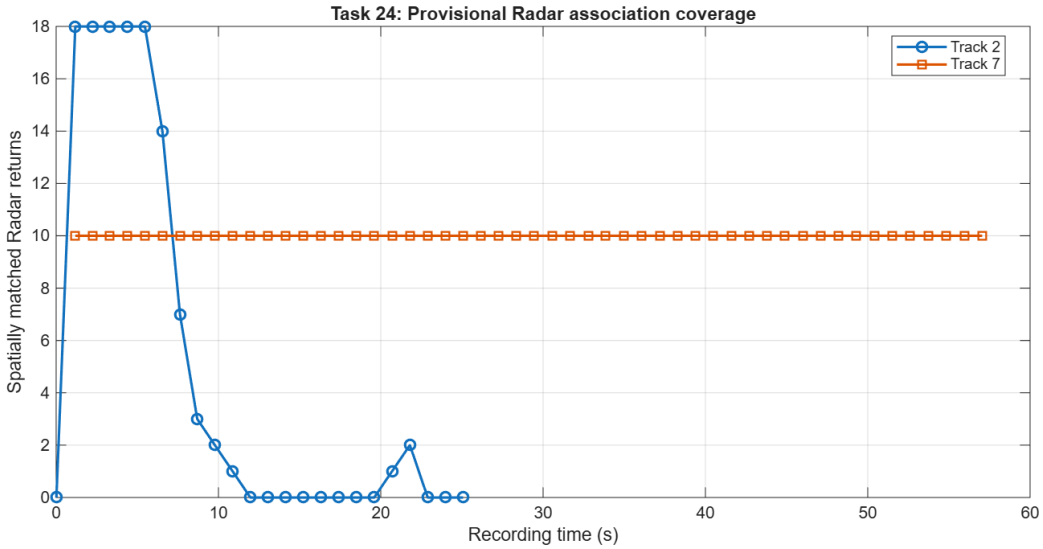
Use in thesis: establishes whether execution load, sensor input volume and target observability should be modeled separately. Radar count alone does not imply association to Track 2.



Original MATLAB Figure 03. Provisional object track trajectories.

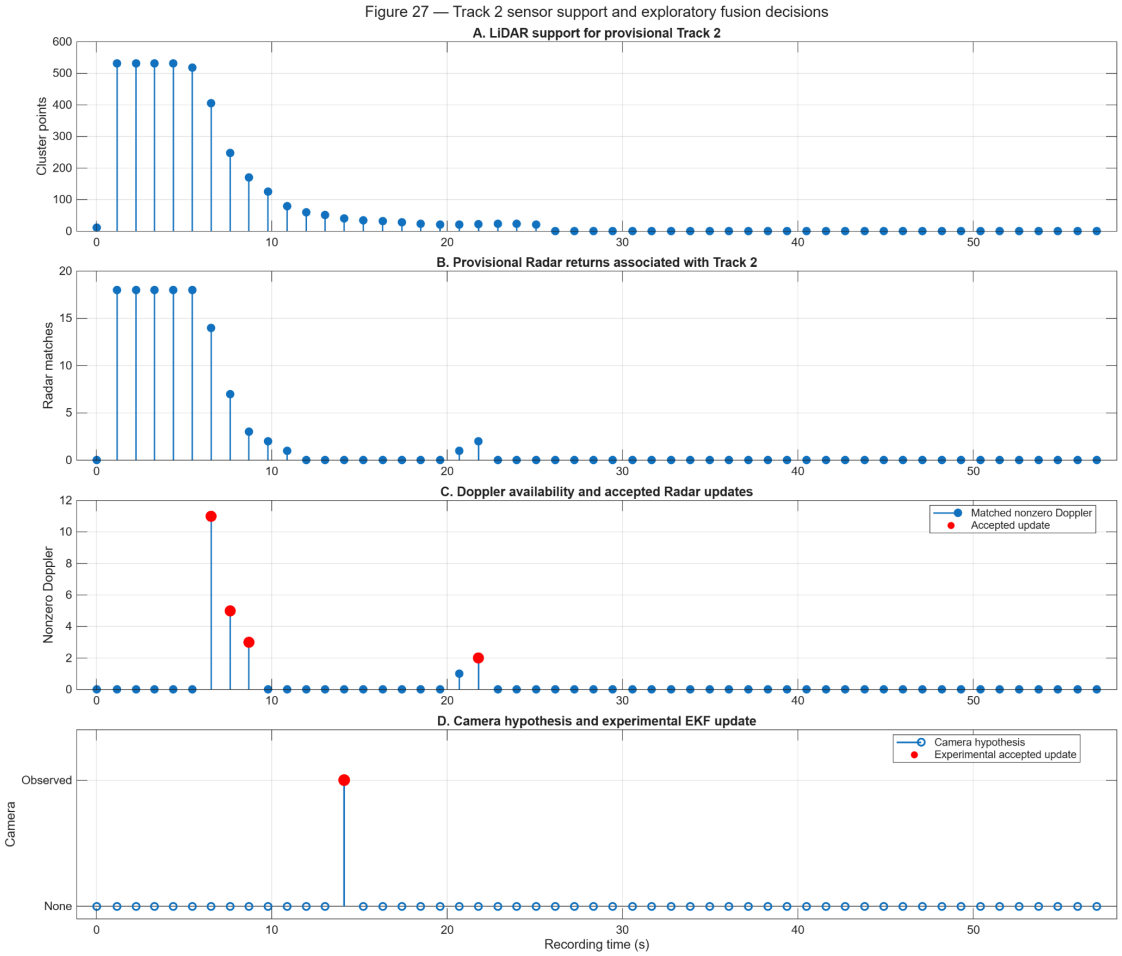
Use in thesis: creates a consistent target ID that can be carried through CPU profiling, FPGA association tests and later reliability studies.

5. Association evidence: when each modality contributes



Original MATLAB Figure 04. Provisional Radar association coverage of Track 2.

Radar returns were available at more times than usable Track 2 associations. This distinction should become an explicit availability-versus-reliability input to Task 25.



Original MATLAB Figure 27. Target-specific sensor support timeline.

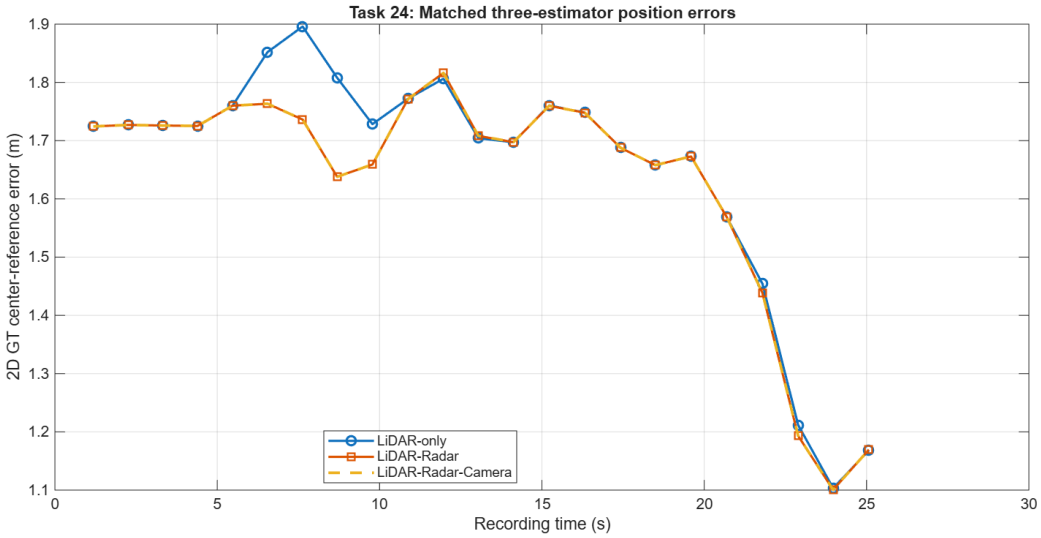
The later absence of Track 2 must not be labeled LiDAR or Radar sensor failure. The tracked object leaves the supported evaluation period while sensors may still produce data.

6. Position-estimation results

6.1 Matched GT-center comparison

Estimator	Mean Euclidean center-reference error
LiDAR-only EKF	1.650417 m
LiDAR-Radar EKF	1.628205 m
Exploratory Camera-LiDAR-Radar EKF	1.628213 m

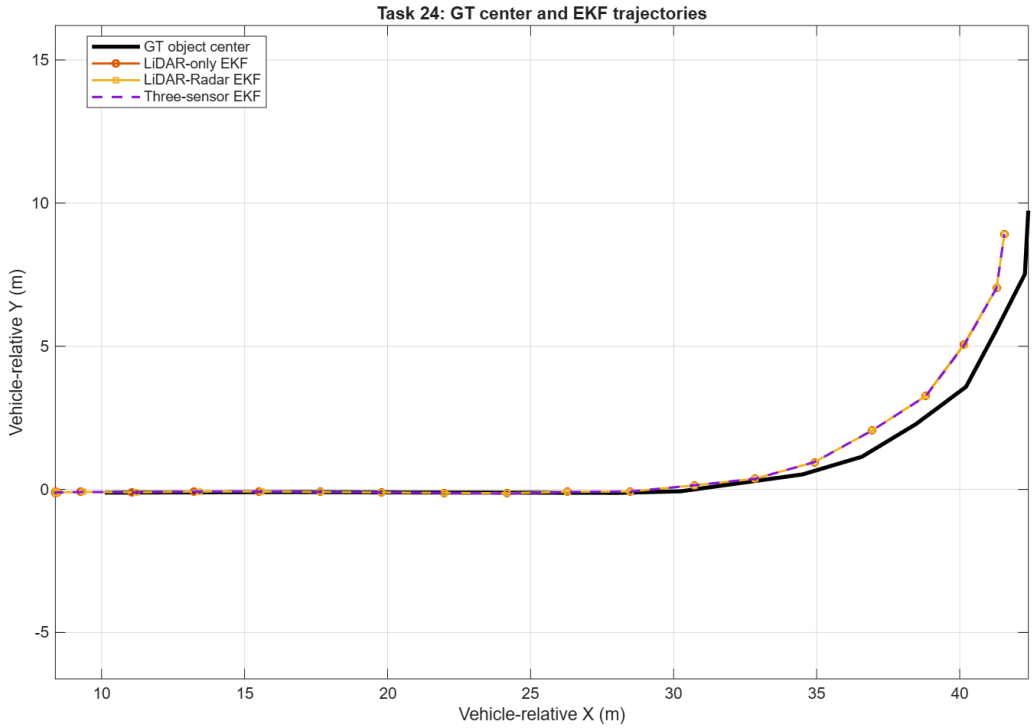
The Radar branch reduced mean GT-center error by 0.022212 m (about 1.35%) compared with the LiDAR-only baseline. The Camera branch was 0.000008 m worse than the Radar branch in this evaluation. These are descriptive within-recording differences, not proof of statistical superiority. Radar improved 11 of 23 samples, worsened six and left six approximately unchanged under the study criterion.



Original MATLAB Figure 06. Matched estimator position errors over time.

The results show mostly overlapping estimators, with localized Radar benefits. A fair hardware benchmark must use the same measurement stream, same association gates and same quality metric.

6.2 Trajectory alignment and systematic bias

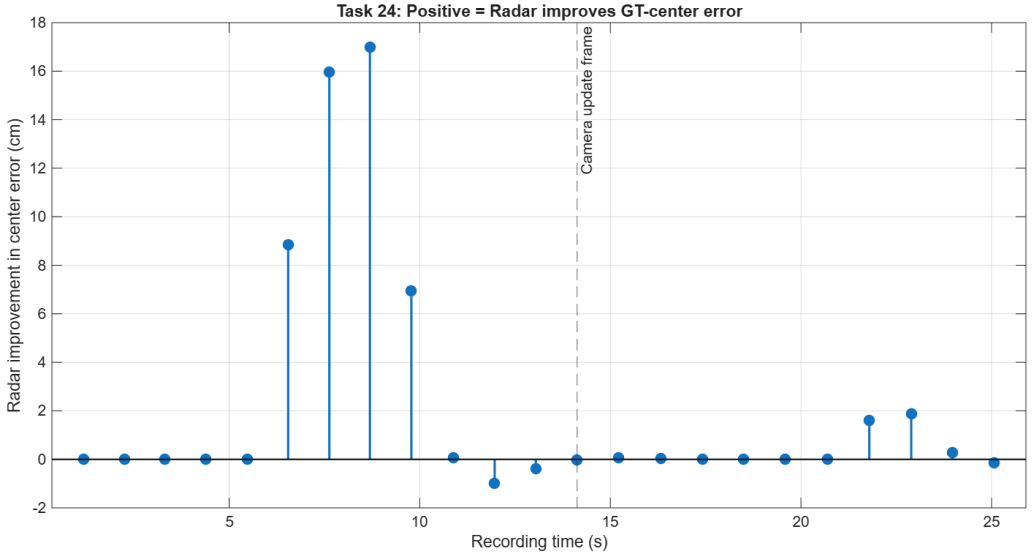


Original MATLAB Figure 07. GT-center trajectory versus LiDAR-only, LiDAR-Radar and exploratory three-sensor estimates.

The overall motion trend is similar, but a persistent physical-reference offset separates LiDAR-observed surface positions from exported GT-center positions.

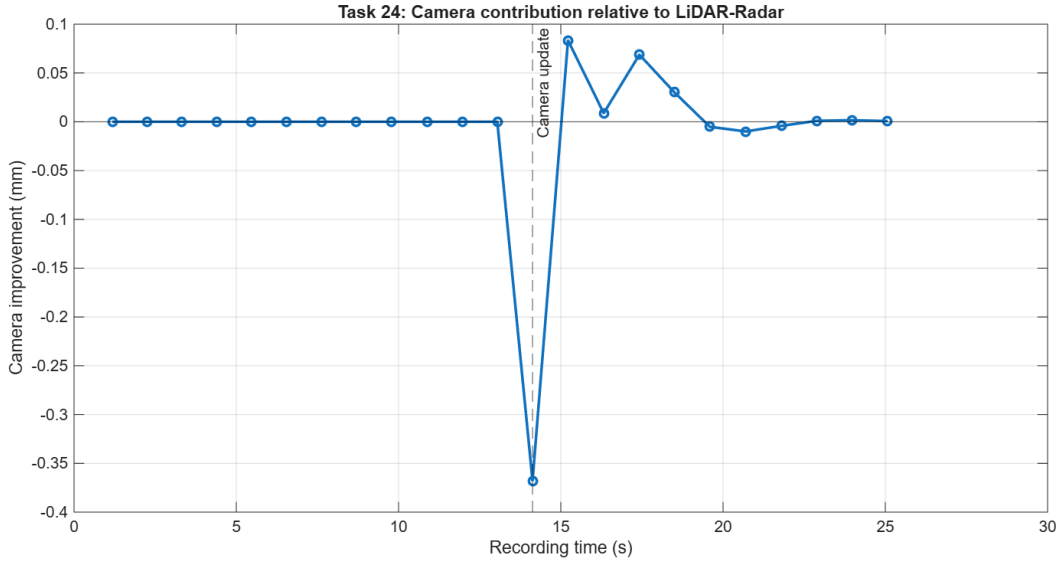
This visual agreement is not a substitute for verifying object reference geometry. A correct surface-associated track can remain consistently displaced from the object center.

6.3 Radar contribution and Camera limitation



Original MATLAB Figure 09. Frame-by-frame change in position error from Radar updates.

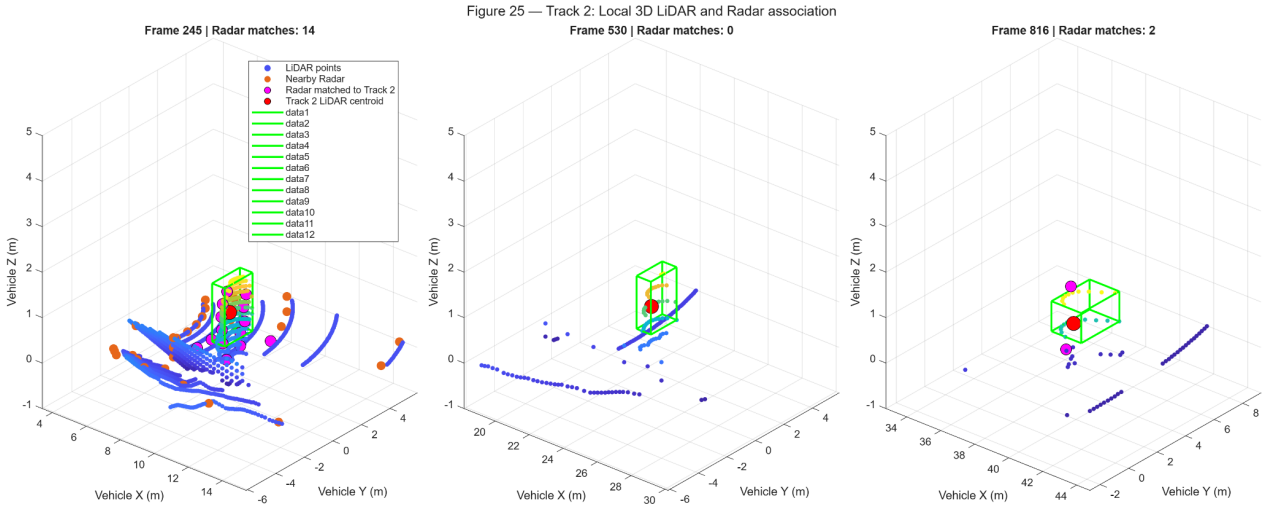
Radar improvements are concentrated at a few moments, which motivates condition-dependent fusion instead of assuming Radar always reduces error.



Original MATLAB Figure 10. Effect of the single exploratory Camera bearing update.

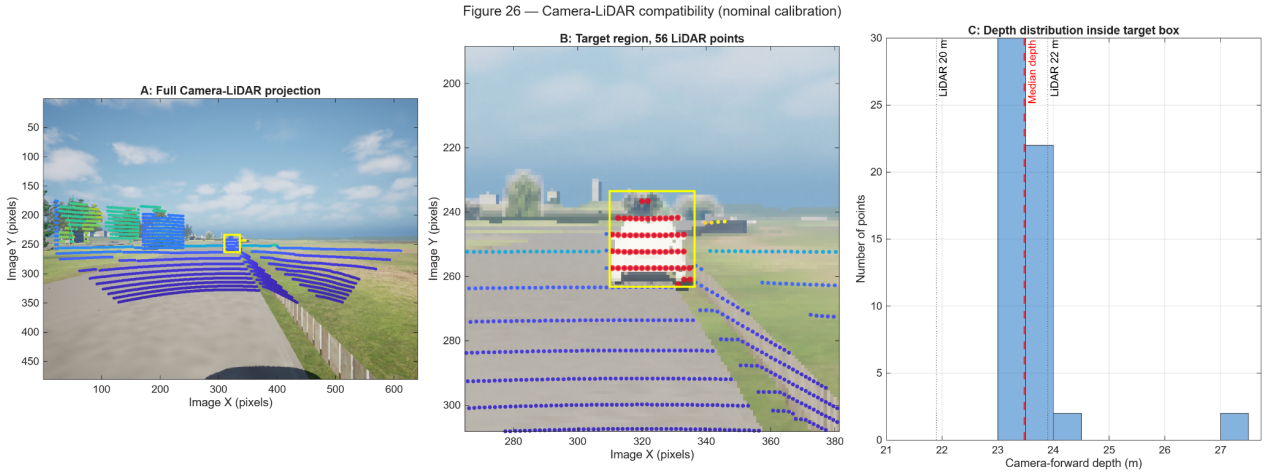
The single Camera measurement at frame 530 did not reduce mean center-reference error; no robust Camera gain can be claimed.

7. Geometric understanding of the scene



Original MATLAB Figure 25. Local 3D LiDAR target geometry and associated Radar returns at frames 245, 530 and 816.

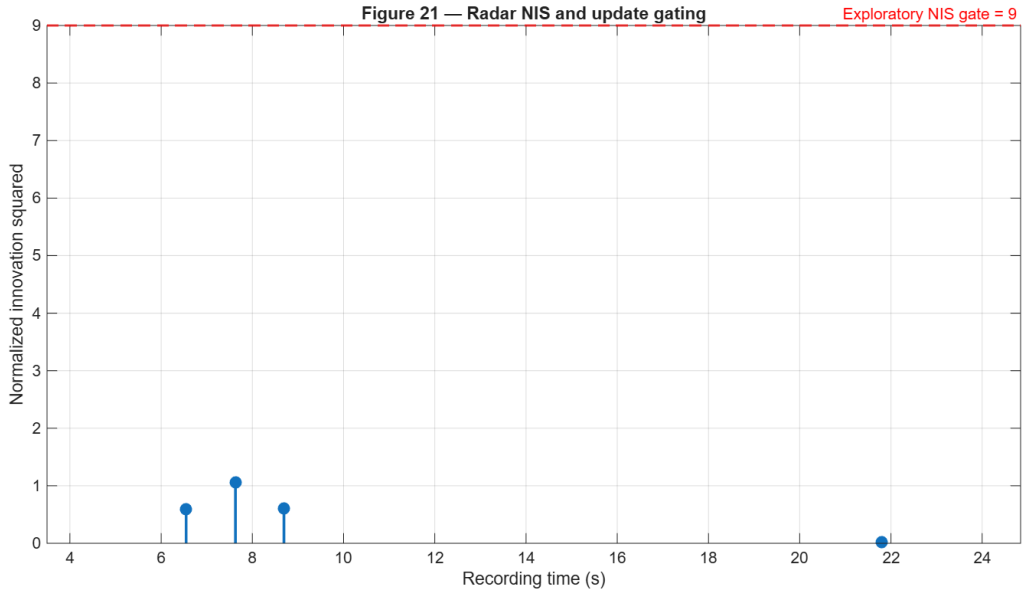
The three examples have 14, 0 and 2 provisional Track 2 Radar matches, respectively. LiDAR cluster boxes visualize geometry, not a certified physical 3D vehicle model.



Original MATLAB Figure 26. Corrected Camera-LiDAR overlay and depth diagnostic (V2).

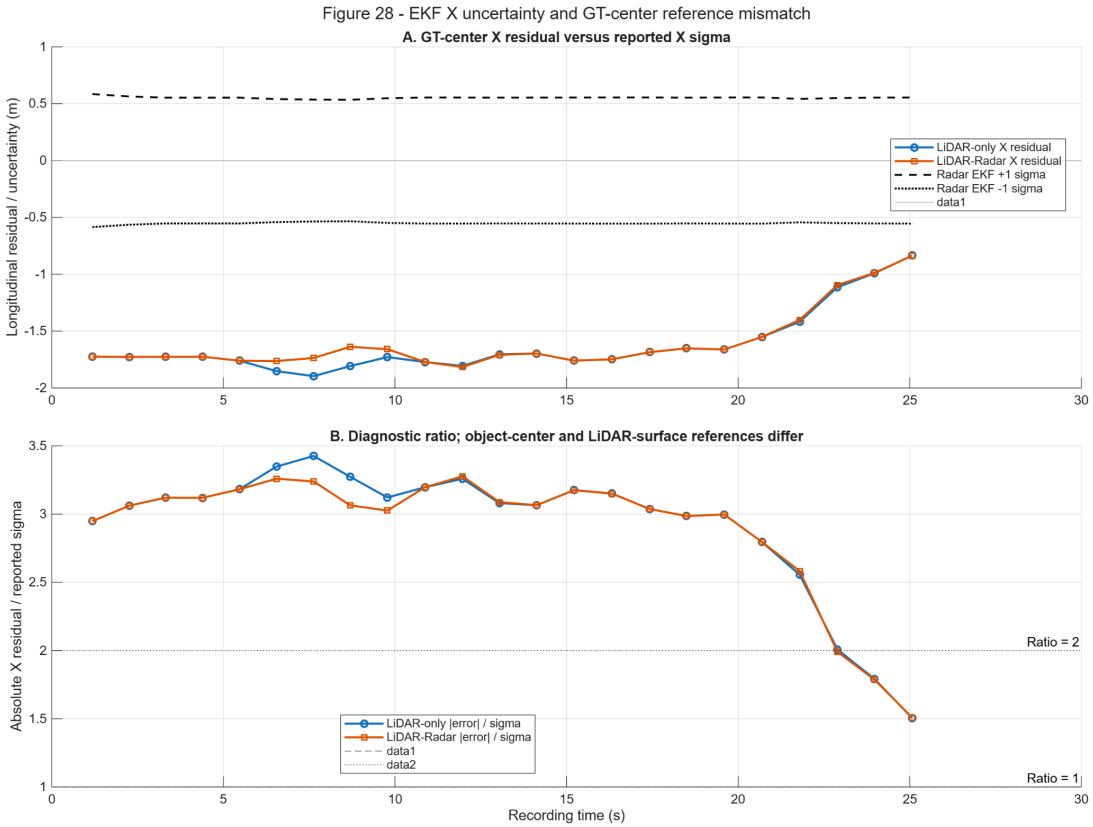
The matched projected-point set supports nominal geometric compatibility at frame 530, but camera calibration and object-class confidence remain insufficient for validated Camera fusion.

8. Innovation gating and estimator uncertainty



Original MATLAB Figure 21. Radar measurement innovation and NIS gating.

Four accepted velocity updates pass the exploratory gate. Gating can support future reliability policies, but requires independent noise calibration before confidence thresholds are treated as meaningful.



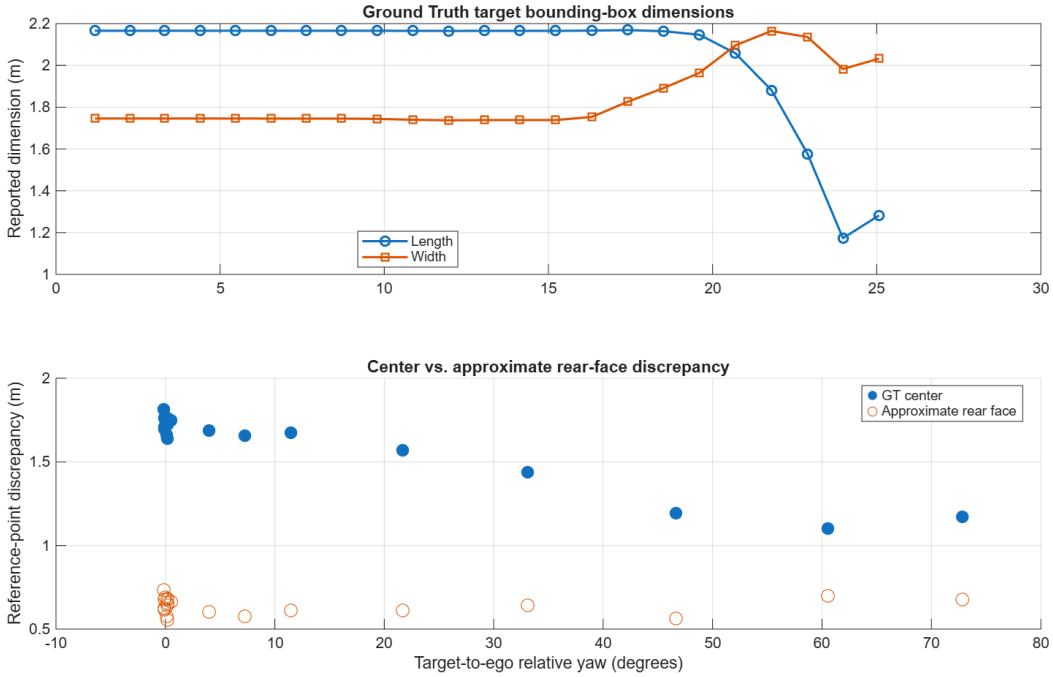
Original MATLAB Figure 28. GT-center X residual and reported EKF X sigma.

The ratio of absolute center-reference residual to reported marginal X uncertainty is typically above 2 early in the sequence. This is not a NEES test because the EKF estimates a LiDAR-surface-associated point, not the GT object center.

9. Ground Truth geometry exporter audit

9.1 Source-confirmed bounding-box representation

The examined ToOSI::toOSIBaseMoving() implementation calls GetActorBounds(false, Origin, Extent) and writes Extent.X, Extent.Y and Extent.Z (each multiplied by 0.01) into OSI dimension.length, width and height. Unreal Extent stores axis-aligned half-extents. Accordingly, the exported values are neither complete physical dimensions nor heading-aligned oriented-box dimensions. The exporter also supplies the actor yaw, creating an inconsistent combination for reconstructing a physical vehicle footprint.



Original MATLAB Figure 12. Reported GT dimensions versus yaw and center/surface reference diagnostics.

The previously used approximate rear-face reference and its about 0.64 m mean diagnostic error must not be reported as validated corrected tracking accuracy, because the underlying dimensions are not oriented full vehicle dimensions.

9.2 Source-confirmed velocity unit problem

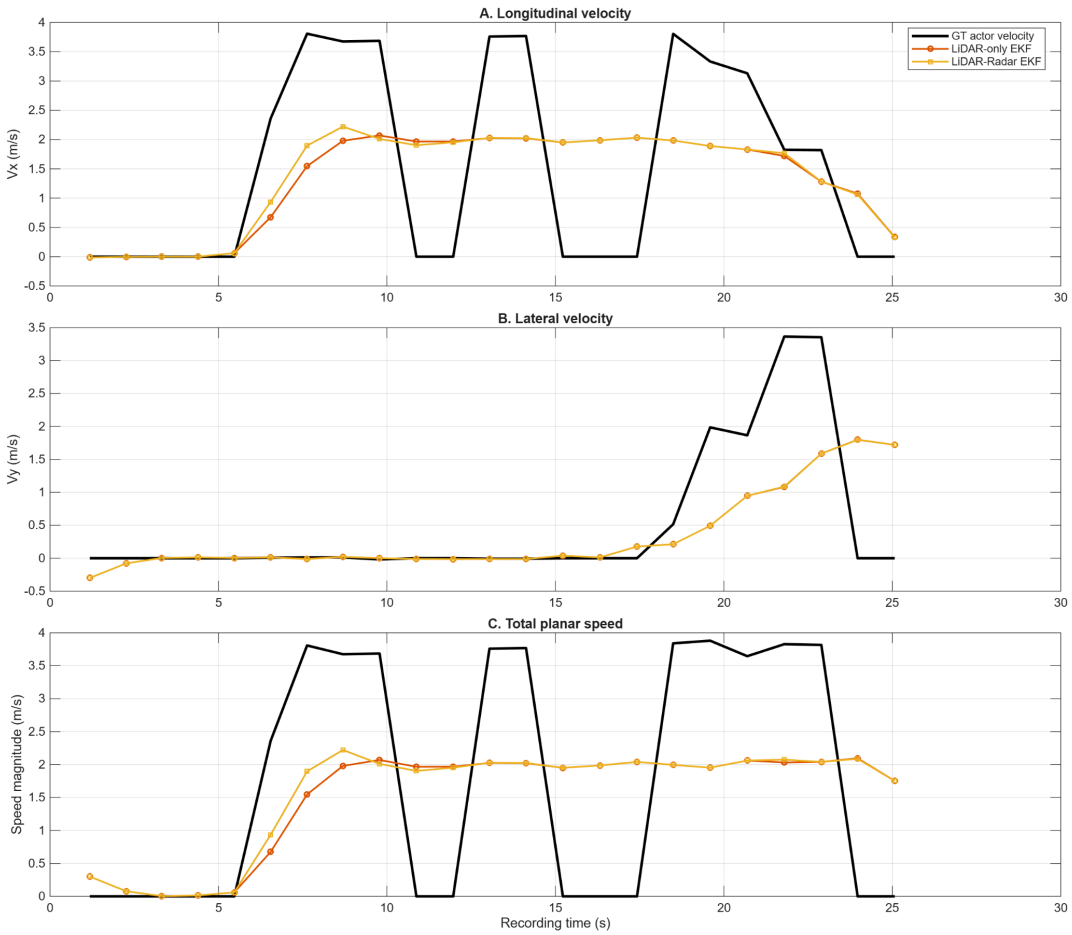
The converter records actor->GetVelocity() directly into OSI velocity fields, without multiplying by 0.01, while position is converted from Unreal centimeters into meters. Under ordinary Unreal world units, these exported velocity components are numerically in cm/s despite the OSI m/s convention. The audit therefore divided raw velocity components by 100 for diagnostic comparison. This unit correction alone did not resolve the full-rate discrepancy.

10. Velocity integrity: central negative result

A 23-frame comparison initially suggested LiDAR-Radar EKF had 1.465717 m/s mean 2D error against exported actor velocity, versus 1.504483 m/s for LiDAR-only. Full-rate validation revealed that this exported actor velocity is not a dependable physical truth reference. These numerical errors are retained as provisional reference-dependent diagnostics, not thesis-level velocity-accuracy claims.

Full-rate GT integrity metric	Result
All source GT messages	2,122 frames
Valid centered position derivatives	2,120 frames
Median logged frame interval	0.026801496 s
Median bounding-center speed	1.863588 m/s
Active samples used for speed ratio	1,028
Median nonzero actor / center speed ratio	1.990366
Mean nonzero actor / center speed ratio	1.989450
Reported zero actor speed with center motion	883 flagged frames

Figure 29 - GT actor velocity and EKF velocity states

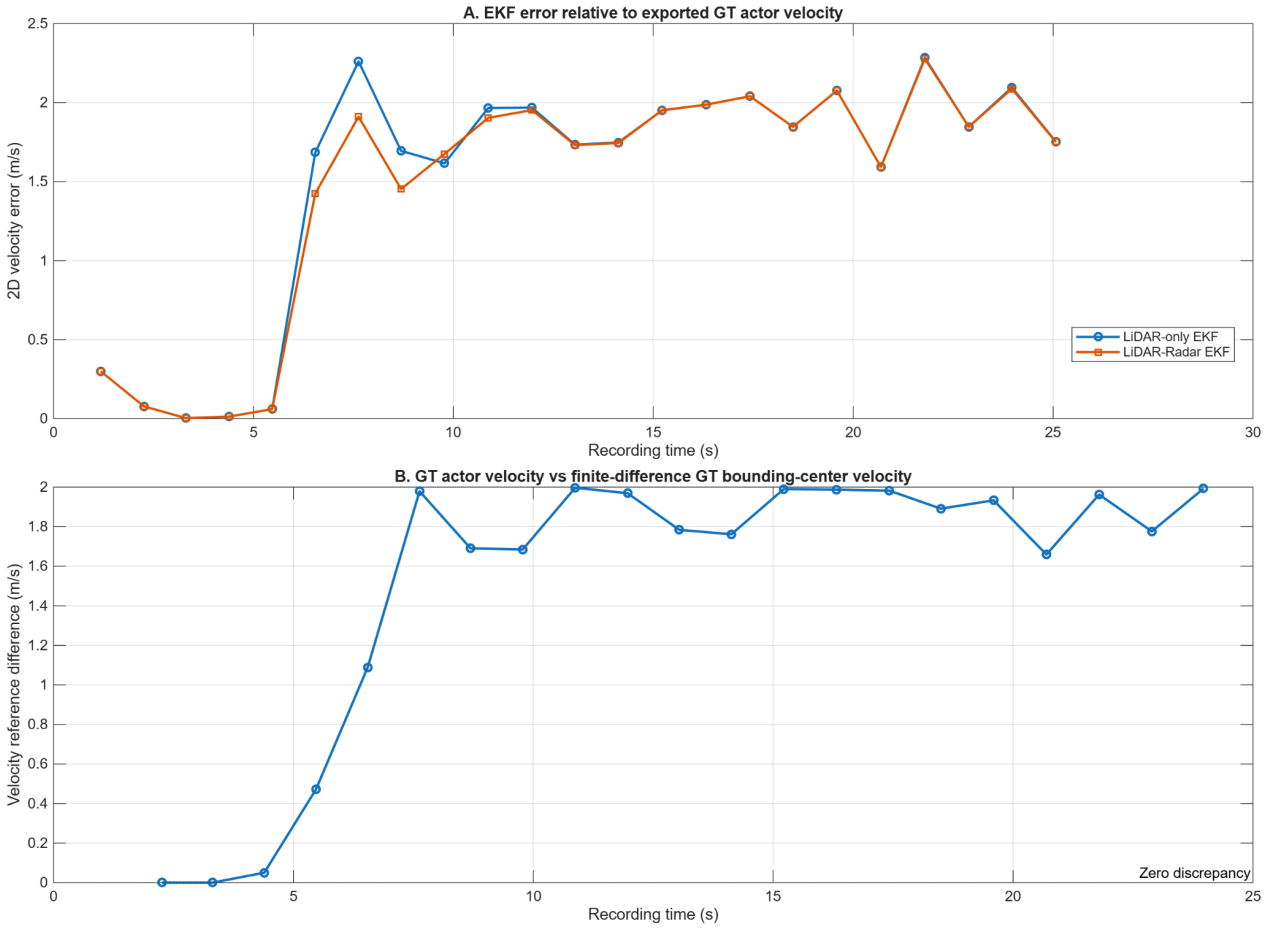


Original MATLAB Figure 29. GT actor velocity versus LiDAR-only and LiDAR-Radar EKF states.

Abrupt GT velocity switches to zero occur amid continuing target displacement. The small Radar velocity advantage is not a validated physical improvement.

10.1 Full-rate disagreement and timing-source inspection

Figure 30 - Velocity error and Ground Truth reference disagreement



Original MATLAB Figure 30. EKF velocity error relative to actor velocity and the actor-versus-position-derived disagreement.

The discrepancy persists in full-rate data. Simply multiplying velocities by 0.5 or substituting filtered values would create an unsupported correction and must not be done.

Source inspection found GT generation in ALogManager::Tick(float DeltaTime), an accumulated logging clock, and actor GetVelocity() sampled by ToOSI. The inspected code does not prove whether the 2x pattern and intermittent zeros arise from actor movement updates, simulation timing, logger sampling phase or another mechanism. Root cause remains unresolved. The ego pose was stationary at all 23 evaluation timestamps; complete intersample ego-motion verification was not performed.

11. Scientific validity and limitations

Issue	Evidence status	Handling in thesis
Position GT correspondence	Strong 23/23 nearest match	Use only for post hoc assessment
LiDAR surface versus object center	Verified reference mismatch	Label center-reference errors; do not claim center-calibrated accuracy
GT dimensions	Exporter half-extents, world-aligned	Do not reconstruct oriented rear/front surfaces without validation
GT actor velocity	Raw unit issue plus intermittent 0 and about 2x discrepancy	Mark velocity comparisons provisional
Camera support	One low-confidence hypothesis, nominal projection	No Camera performance claim
Radar Doppler/NIS	Four accepted updates, assumed noise	No calibrated gating/NEES claim
External validity	Single scene, 23 correlated GT samples	Require independent scenes and repeats
Energy / latency / FPGA	No verified measurements in Task 24	Must be measured in later tasks

What the experiment does and does not demonstrate

Demonstrates: traceable multi-sensor preprocessing and candidate association; a reproducible LiDAR-centered EKF baseline; conditional Radar velocity updates; an explicitly experimental Camera update; independent Ground Truth correspondence; and source-verified data integrity limitations.

Does not demonstrate: verified physical vehicle-center accuracy, validated velocity RMSE, safe operation across environments, full-rate real-time execution, robust three-way sensor association, or an energy saving from FPGA acceleration.

12. Contributions to the thesis

C1 - Measurement-aware fusion foundation. Establishes concrete preprocessed inputs, association rules and state-update kernels for implementation-level profiling. The CPU baseline can be frozen and reused bit-for-bit (or within defined numerical tolerance) for hardware comparisons.

C2 - Reliability factors grounded in observations. Separates sensor stream availability, target visibility, association support, Doppler validity, Camera confidence and innovation acceptance. This directly informs Task 25 reliability features.

C3 - Honest quality constraint. Exposes why comparing a LiDAR surface estimate against an actor-derived GT center can overstate filter errors. A consistent target reference and independent quality protocol are prerequisites for claiming energy per valid perception result.

C4 - Integrity of reference data. The OSI geometry and velocity audit prevents deployment decisions and hardware tradeoffs from being optimized against inconsistent or mislabeled references.

13. Recommendations and next-task acceptance criteria

Stage	Required action	Pass criterion
Task 25	Specify measurable target observability and confidence, calibrate measurement noise and define fallback behavior	Document reliability decisions separately from sensor presence
Task 26	Profile fixed CPU kernels with identical sensor inputs, quality gates and track semantics	Reproducible mean/p95/p99 timing, workload and memory traces
Tasks 27-29	Select candidate accelerators by computational and data-transfer costs	Baseline design-space comparison without invented energy results
Task 30+	Deploy ARM-FPGA implementation and measure whole-platform power	Measured energy per valid result and quality/p99 latency constraints
Task 42+	Integrate continuous clear/fog/traffic and decision telemetry	Matched live comparison across randomized/repeated scenarios

Consolidation acceptance checklist for Task 24.11

The user inventory reports 365 research files: 59 Python scripts, 18 MATLAB scripts, 99 CSVs, 100 PNGs, 39 TXT files, nine Markdown files, eight INI files, four JSON files and additional formats. All 30 MATLAB figure files, all nine named report sections and the 30-row figure index were found. Fifteen duplicate filename groups exist; they are not errors in themselves. Preserve full relative paths and generate a SHA-256 manifest. Never flatten same-named artifacts into one destination folder or silently overwrite them.

The live Windows E: drive cannot be directly inspected from this PDF-generating environment. Counts are taken from the supplied PowerShell inventory; an actual file-by-file checksum audit must still run locally during consolidation. The research report is a synthesis of supplied results, not an independent rerun of their EKF calculations.

14. Overall conclusion

Task 24 succeeded as a controlled, auditable feasibility study for multi-sensor tracking. Radar offered a modest improvement in one development recording; Camera contributed an exploratory bearing hypothesis but no demonstrated positional benefit. More importantly, a reliable pathway from raw OSI streams through association, EKF updates, evaluation and reference-data auditing was established. This is the appropriate foundation for the thesis energy-efficiency question: only once quality is defined and validated can ARM CPU versus FPGA scheduling and power claims be assessed meaningfully. Task 25 should improve the reliability model; Tasks 26 onward must measure execution cost and ultimately total system energy.

Appendix A. Complete original MATLAB figure inventory

Figure	Subject	Artifact filename
01	Input availability	Task24_Figure_01.png
02	LiDAR filtering and clustering	Task24_Figure_02.png
03	Provisional tracks	Task24_Figure_03.png
04	Radar association	Task24_Figure_04.png
05	Radar-LiDAR velocity discrepancy	Task24_Figure_05.png
06	GT position errors	Task24_Figure_06.png
07	GT/EKF trajectory	Task24_Figure_07.png
08	Signed position bias	Task24_Figure_08.png
09	Radar error improvement	Task24_Figure_09.png
10	Camera effect	Task24_Figure_10.png

Figure	Subject	Artifact filename
11	Position error CDF	Task24_Figure_11.png
12	GT dimension/yaw diagnostics	Task24_Figure_12.png
13	3D LiDAR scene	Task24_Figure_13.png
14	3D Radar scene	Task24_Figure_14.png
15	Camera-LiDAR image projection	Task24_Figure_15.png
16	Combined 3D geometry	Task24_Figure_16.png
17	3D track progression	Task24_Figure_17.png
18	Three-frame geometry	Task24_Figure_18.png
19	EKF estimated velocities	Task24_Figure_19.png
20	Radar measured/predicted velocity	Task24_Figure_20.png

Figure	Subject	Artifact filename
21	NIS gating	Task24_Figure_21.png
22	EKF X uncertainty	Task24_Figure_22.png
23	Estimator separation	Task24_Figure_23.png
24	Radar update effects	Task24_Figure_24.png
25	Target-local 3D associations	Task24_Figure_25.png
26	Corrected camera-depth projection	Task24_Figure_26_V2.png

Figure	Subject	Artifact filename
27	Sensor support timeline	Task24_Figure_27.png
28	GT-center residual versus sigma	Task24_Figure_28.png
29	GT actor versus EKF velocity	Task24_Figure_29.png
30	Velocity integrity diagnostics	Task24_Figure_30.png

Appendix B. Reproducibility and source record

Source trace: clean C02 V2 recordings 20261108T211139Z_sv_380_X_2122_GEMSTAR.osi and 20261108T211140Z_gt_380_X_2122_GEMSTAR.osi; synchronized 2,122-message pair. Research outputs maintained under tests/Task_24_Final_Research/09_Controlled_Scenario/C02_MovingTarget. Original MATLAB files Task24_Figure_01.png through Task24_Figure_30.png (with corrected Task24_Figure_26_V2.png) are embedded as selected illustrative evidence.

Local audit inputs: gt_center_comparison_track2_v1.csv; gt_track2_identity_audit_v1.csv; gt_ekf_velocity_comparison_v1.csv; ego_gt_velocity_integrity_audit_v1.csv; gt_fullrate_velocity_audit_v1.csv; Task24_MATLAB_30_Figure_Index_V1.csv; Task24_Complete_MATLAB_Research_Report_V1.md.

Source inspection: ToOSI.cpp (toOSIBaseMoving), OSI_GroundTruth.cpp (parseWorldToGroundTruth and timestamp), LogManager.cpp (Tick and serialization). Observations based on the supplied copied source excerpts; compiler/runtime behavior beyond those snippets is not independently verified.

Interpretation convention: “verified” means supported by the supplied local command output or inspected code, not externally reproduced. All energy/safety and generalization claims are deferred until independent future benchmarks.